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Detecting cyber employment fraud in platform-based space: A multi-agent large language model approach

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ABSTRACT

Cyber employment fraud causes more than $2.6 billion in annual losses and severe psychological harm to job seekers, yet law enforcement struggles to combat these platform-mediated offenses. Although machine learning techniques are increasingly used to detect cyber employment fraud, the potential of large language models (LLMs) remains largely unexplored, particularly in relation to the geographically distributed nature of online labor markets and platform ecosystems. This study examines whether ChatGPT-4o can effectively identify fraudulent job postings using the Employment Scam Aegean Dataset (EMSCAD). We introduce a single-pass ChatGPT-4o API call and a multi-agent framework featuring a specialized judging agent and three sub-agents for consistency verification, legitimacy assessment, and contextual validation with QCEW database integration. Both approaches generate fraud risk scores (0–10 scale), which are integrated as features into Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), and Multilayer Perceptron Neural Network (MLP) models using Monte Carlo cross-validation. Results show that both approaches can significantly improve machine learning model performance metrics. These ChatGPT-4o generated risk scores exhibit a statistically significant monotonic relationship with fraudulent prevalence rates, indicating its feasibility for being integrated into fraud detection applications and early warning systems across platforms and jurisdictions. Overall, the study serves as proof of concept for integrating LLMs in cybercrime prevention and criminal justice practice, demonstrating that both single-pass approach and multi-agent framework have unique advantages and potential in benefiting cyber employment fraud prevention in platform-based space.

1. Introduction

Over the past two decades, the dramatic growth of internet users has been fundamentally transforming economic activity and reshaping crime opportunities worldwide (Stackscale, 2023). While traditional street crime has declined in many countries, cybercrime has increased rapidly, creating new challenges for criminal justice systems (U.S. Crime Rates and Trends—Analysis of FBI Crime Statistics | Brennan Center for Justice, 2026; Magazine, 2021). Cybercrime, defined as offenses conducted through computers and internet enabled systems (ICE, 2025; Onwuadiamu, 2025; Wall, 2003), now generates global losses in the trillions of dollars, with continued growth projected (Magazine, 2021). These developments raise critical questions for criminology regarding how offending, victimization, and prevention operate in digitally mediated environments.

Cyber employment fraud has become one of the most prevalent and costly forms of online fraud (FBI, 2025; Gong et al., 2025). Offenders impersonate legitimate organizations and post fraudulent job advertisements on online platforms, inducing victims to disclose sensitive information, transfer funds, or participate in illicit financial transactions such as cryptocurrency-based money laundering (FBI, 2023; FBI, 2025; Sarkar & Shukla, 2023). In 2024 alone, the FBI documented 20,044 incidents of cyber employment fraud, resulting in more than $2.6 billion in financial losses (FBI, 2025). These schemes illustrate how digital platforms function as criminogenic environments in which motivated offenders, suitable targets, and weak guardianship converge.

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The consequences of cyber employment fraud extend beyond financial loss to include substantial psychological harm. Victims often report shame, anxiety, and diminished trust in institutions and labor markets, with younger job seekers particularly vulnerable due to limited experience and heavy reliance on online job searches (Cazanis et al., 2025; Szymkowiak et al., 2021). Despite recent regulatory efforts, such as the Federal Trade Commission's 2024 impersonation rule and repeated warnings issued by the FBI (FBI, 2024; FBI, 2025), cyber employment fraud remains difficult to detect and prevent. The global and transient nature of online platforms complicates traditional criminal justice practices related to guardianship, evidence retention, and offender identification (Calderón & Miller, 2020; Doig et al., 2024; Koziarski & Lee, 2020; Sarkar & Shukla, 2023; Willits & Nowacki, 2016). As a result, proactive and technology enabled forms of guardianship are increasingly necessary to mitigate risks in digital environments.

This study responds to these challenges by examining whether ChatGPT-4o can detect fraudulent job postings using the Employment Scam Aegean Dataset (EMSCAD). We compare a multiagent large language model (LLM) framework with a single-pass LLM API call evaluation approach. The study addresses three research questions: (1) How effectively do LLM-based approaches detect fraudulent job postings?; (2) How does single-pass API call approach compare with multi-agent framework system?; and (3) How do domain knowledge informed verification processes influence performance? By conceptualizing LLMs as capable guardians in cyberspace, this research extends traditional crime prevention frameworks and contributes to both criminal justice scholarship and policy-oriented strategies for cybercrime detection and prevention.

2. Literature review

The prevalence of employment fraud in platform-based space is rooted in Routine Activities Theory, which suggests that the convergence of motivated offenders, suitable targets, and weak guardianship elevates the risk of crime (Leukfeldt & Yar, 2016). In the digital platform-based space, this is reflected in the lower cost of offending and expanded access to suitable targets. Furthermore, conventional crime prevention that heavily relies on human patrollers and reactive policing cannot be deployed at the required scale in virtual environments due to the sheer volume of real time online traffic. For example, employment platforms such as Indeed and LinkedIn add approximately 10 new job postings each second (PitchMeAI Editorial Team, 2026), of which only a small fraction may be posted by fraudsters.

Drawing on situational crime prevention principles, several studies have explored automated forms of guardianship capable of identifying suspicious postings before victims are exposed (Gong et al., 2025; Kolupuri et al., 2025). More recently, the rapid deployments of LLMs and Agentic AI create both opportunities and threats to counter cyber employment frauds (Finistrella et al., 2025). These developments are reviewed in detail in Sections 2.1 and 2.2.

Beyond Routine Activity Theory and situational crime prevention principles, this study employs faking and strategic response distortion literature to bring a complementary psychological perspective on fraudsters' behavioral strategies. Faking refers to motivated information distortion in which actors strategically manipulate identity, memory, judgment or behavior to achieve instrumental goals (Bensch et al., 2019). In the context of cyber employment fraud, offenders deliberately manipulate organizational identity, compensation, urgency, and use pressure tactics to deceive job seekers. According to taxonomy of faking behaviors, the construction of fraudulent job postings can be analyzed by identifying faking signals at different stages (comprehension, retrieval, judgment, and response) (Röhner et al., 2025). The retrospective alignment between faking taxonomy and the multi-agent system design is discussed in Sections 3.3.2 and 6.1.

Cybercrime prevention is inherently interdisciplinary, involving criminology, computer science, psychology, sociology, and information systems. Over-reliance on technical tools could lead to "automation bias", which creates new vulnerabilities by reducing critical analysis (Butavicius et al., 2020; Tilbury et al., 2026). Effective cybercrime research requires integrating criminological domain knowledge and contextual understanding into predictive frameworks (Nanath & Olney, 2023). Relevant previous efforts are introduced in Section 2.3.

2.1. Machine learning models for cyber fraud detection

Prior studies have used open data and social media data to train ML and deep learning (DL) models for automated cyber fraud detection, achieving 70% to 90% accuracy rates (Kolupuri et al., 2025; Kumari et al., 2018; Mahor et al., 2021). Specifically, in the cyber employment fraud detection domain, many studies used the publicly available Employment Scam Aegean Dataset (EMSCAD) created by Vidros et al. (2017). These studies tested a wide range of ML and DL models, including Logistic Regression (Nandhini & Marseline, 2020), Random Forest (Sofy et al., 2023), K Nearest Neighbors (Habiba et al., 2021), Multilayer Perceptron (Habiba et al., 2021), LightGBM (Tabassum et al., 2021), Naïve Bayes (Mahbub & Pardede, 2018), Gradient Boosting (Naudé et al., 2023), XGBoost (Gong et al., 2025), and Deep Neural Networks (Akram et al., 2024). While many models in these studies achieved good performance, substantial opportunities for further improvements remain. The EMSCAD dataset and cyber employment fraud postings in general are text heavy, requiring natural language processing (NLP) methods to convert text into numerical representations (Gong et al., 2025). Such efforts typically emphasize the occurrence and frequency of specific words and word combinations, rather than comprehensively evaluating the semantic and contextual meanings of the texts as humans would. The high dimensionality and complexity of unstructured text data often lead to missed detection of sophisticated fraud patterns (Jacob et al., 2025; Tong & Shen, 2023).

In addition, fraudulent postings constitute relative minority cases compared to legitimate ones in both EMSCAD dataset and most online employment platforms (Vidros et al., 2017). The class imbalance can cause ML models to be biased toward the majority cases, resulting in an increased likelihood of missing fraudulent activities (Qamar et al., 2019; Xu et al., 2023). A high number of false negative predictions will increase operational costs and erode operator confidence (Alhasnawi et al., 2025).

2.2. Trends in large language models (LLMs)

LLMs are trained on vast language data to produce human-like responses, with capabilities for understanding both the meanings and contextual relationships within text (Vaswani et al., 2017). Recent evaluations demonstrated that LLMs possess advanced reasoning and common-sense understanding capabilities comparable to human cognition (Dentella et al., 2024; Schmitz & Sonnleitner, 2025; Xiao et al., 2023). Growing applications of LLMs as communication, planning, and task monitoring assistants have emerged in entertainment, research, and healthcare domains (Amjad et al., 2023; Li et al., 2025; Peng et al., 2023).

The application of LLMs to the field of criminology is relatively recent. Emerging studies have preliminary focused on judicial decision-making (Liu & Li, 2024; Samee et al., 2024), police investigations and reporting (Adams, 2024; Dement & Inglis, 2024; Kim et al., 2024). Some studies have begun exploring the applications of LLMs in cybersecurity. For example, researchers evaluated ChatGPT-3.5 and GPT-4 performances on identifying linguistic and structural indicators of phishing emails (Jiang, 2024). Hassanin and Moustafa reviewed applications of LLMs in cybersecurity, covering intrusion detection, email phishing prevention, and LLMs-based Honeypots, while also acknowledging the increasing use of LLMs in developing more cyberattacks (Hassanin & Moustafa, 2024). Despite these efforts, exploration of any potential

applications of LLMs to cybercrime, particularly for cyber employment fraud detection and prevention domain, remains limited.

Another prominent trend in LLM research is the growing attention to multi-agent frameworks, which enable collaborative intelligence and efficient task execution to enhance cybersecurity (Finistrella et al., 2025; Wang & Li, 2025). Multi-agent frameworks involve multiple independent agents working collaboratively, each specializing in different aspects of analysis or employing distinct methodologies, which allows breaking down complex problems into specialized tasks (illustrated in Fig. 1) (Hanna et al., 2025). The task specialization provides multi-agent frameworks advantages in analyzing long and complex cyber employment fraud text data by improving factual accuracy and reducing bias (Albashrawi, 2025; Almohaimeed & Azmi, 2025; Trinidad et al., 2025). Still, using multi-agent frameworks to understand cyber employment fraud prevention appears underexplored.

2.3. Domain knowledge and feature engineering in fraud detection

Feature engineering, which involves transforming raw data into more informative features, has been shown to substantially improve ML model performance in crime detection (Da Silva et al., 2020; Gong et al., 2025; Santos et al., 2024; Santos Junior et al., 2021). Feature engineering heavily relies on human knowledge and experience in the domain (Deng et al., 2020). For example, in credit card fraud detection, indicators such as customer spending behaviors, transaction timing and frequencies are found to have strong implications for fraud (Zhang et al., 2021). These features are often identified through an iterative process of feature engineering and model evaluation (Deng et al., 2020).

Feature engineering also increases the interpretability of models, which allows experts to understand why an online employment posting is flagged as suspicious and facilitating subsequent investigations (Deng et al., 2020). Advanced ML models, particularly DL models, often lack interpretability due to their complex architecture and non-linearity (Chakraborty et al., 2017). Domain knowledge informed feature engineering can improve model interpretability by revealing patterns in unstructured, high dimensional text data (Baesens et al., 2021; Rao et al., 2025).

Despite its benefits, the iterative process of feature engineering can be complex and resource intensive (Lucas et al., 2020). To address this challenge, Nanath and Olney (Nanath & Olney, 2023) adopted a crowdsourcing technique to collect human experts' perceived risk assessments on a subset of job postings in the EMSCAD dataset. These features were used to train traditional ML models, where improvements in key performance metrics were observed. Still, this technique requires intensive human labor to manually review and label each entry in the dataset, which hinders its adoption in the industry. To the best of our knowledge, no study has explored the potential of LLMs in rule-based feature engineering in the domain of cyber employment fraud prevention.

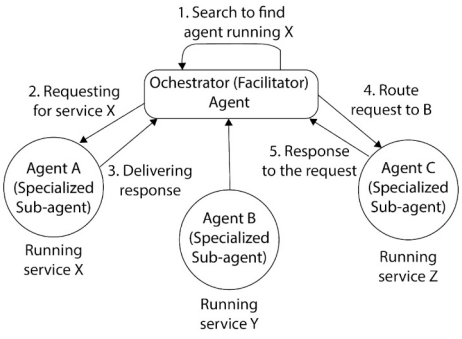


Fig. 1. Implementation of a sample multi-agent system. Source: (Dorri et al., 2018). Graphically adapted by the authors.

3. Data and methodology

3.1. Research design

Nanath and Olney (2023a) demonstrated that crowdsourced human risk perceptions can enhance ML models for online recruitment fraud detection. Building on this approach, the present study examines whether LLM-generated risk perceptions can similarly improve fraud classification performance. As illustrated in Fig. 2, the research design integrates structured job posting data with LLM-derived risk scores to construct a baseline dataset and two experimental variants. In the baseline condition, models are trained solely in structured job posting features. The first experimental condition augments these features with a single risk score generated via a constrained LLM prompt, while the second incorporates a risk score produced by a role-specialized, multi-agent LLM framework. Model performance across all three datasets is evaluated using Monte-Carlo Cross Validation.

This design aims to assess the conceptual value of LLM-generated perceptual features in employment fraud detection rather than compare prompt engineering strategies. Detailed descriptions of datasets, preprocessing, feature engineering, model selection, and evaluation procedures follow in subsequent sections.

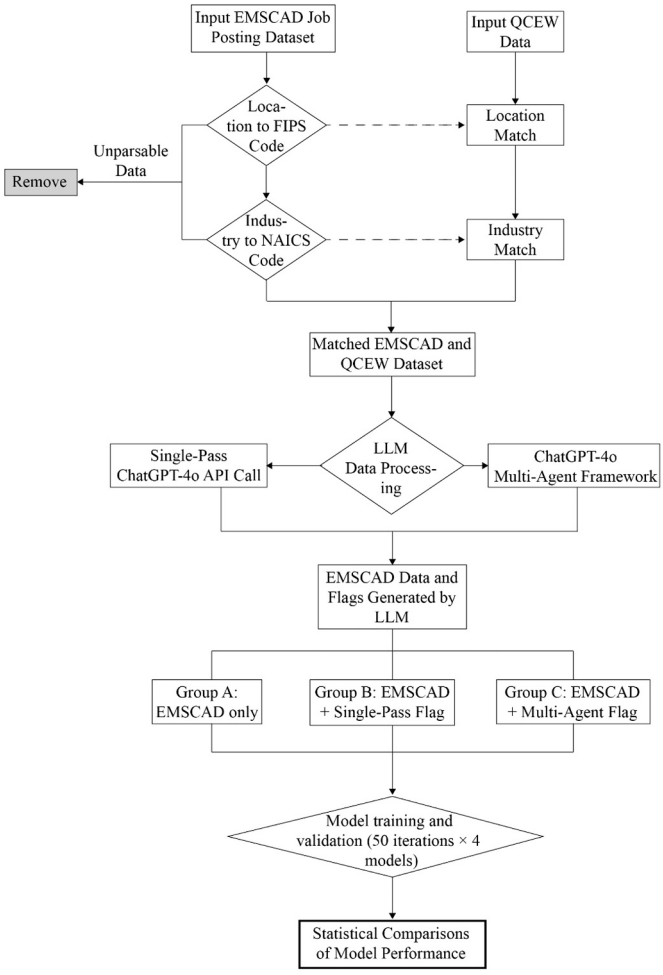


Fig. 2. Overview of the research workflow. EMSCAD job postings are matched to QCEW data via FIPS and NAICS codes, then evaluated by ChatGPT-4o using single-pass API call and multi-agent framework to generate risk scores and fraud indicator flags. The EMSCAD only baseline machine learning models are compared against single-pass API call and multi-agent framework risk scores enhanced ones by using Monte Carlo cross-validation.

3.2. Datasets

The primary dataset used in this study is the Employment Scam Aegean Dataset (EMSCAD), a real-world collection of online job postings obtained from an Applicant Tracking System between 2012 and 2014 (Vidros et al., 2017). EMSCAD contains 17,880 job advertisements worldwide, of which 866 are verified as fraudulent (4.8%), and includes both structured metadata (e.g., location, industry, employment type) and unstructured text fields (e.g., job descriptions and company profiles), along with a binary fraud label (Gong et al., 2025). Since its release, EMSCAD has become a benchmark dataset for NLP and ML research on online employment fraud detection (Gong et al., 2025; Mahbub & Pardede, 2018; Nanath & Olney, 2023; Nindyati & Bagus Baskara Nugraha, 2019). Its widespread adoption enables direct comparison of our multi-agent framework's performance against established baselines and facilitates reproducibility across studies. Despite the temporal limitations of the EMSCAD dataset and that specific fraud tactics may evolve, the fundamental deceptive mechanisms underlying employment fraud, such as information inconsistency, unrealistic compensation, requests for sensitive information, and pressure tactics, represent persistent characteristics of fraudulent behavior that transcend technological eras.

To provide contextual information, the study integrates county-level macroeconomic data from the 2013 Quarterly Census of Employment and Wages (QCEW), maintained by the U.S. Bureau of Labor Statistics (Bureau of Labor Statistics, 2026; Button, 2019; Nguyen, 2024). QCEW reports county level employment and average annual wages by industry, which was stored locally as an external knowledge base accessible to the AI agents.

Dataset integration relied on supplementary resources, including the 2012 North American Industry Classification System (NAICS) index files and the United States Counties Database (U.S. Census Bureau, 2026; US Counties Database | Simplemaps.com, 2026), enabling alignment between EMSCAD job postings and corresponding geographic and industry-level economic indicators.

3.3. Data processing

3.3.1. Data integration

The context-aware AI agent integrates county-level economic indicators from the QCEW dataset by matching EMSCAD job postings employers' business establishment location and business type to corresponding geographic and industry records. EMSCAD location fields were mapped to QCEW county FIPS codes using the SimpleMaps U.S. Counties Database, with non-U.S. or unmappable entries excluded.

Industry matching posed greater challenges due to free-text EMSCAD entries and standardized NAICS codes in QCEW. A four-step process was implemented: (1) exact string matching, (2) fuzzy matching, (3) manual semantic mapping, and (4) LLM-assisted inference. Exact and fuzzy matching yielded 1.07% and 12.7% of matches, respectively. Manual semantic mapping using the NAICS Code Search tool accounted for the majority of matches (72.58%), while LLM-assisted inference assigned NAICS sectors to entries with missing industry values, with low-confidence predictions manually reviewed.

The final integrated dataset consisted of 8781 EMSCAD entries (556 fraudulent), each linked to QCEW contextual indicators including average annual wages, employment levels, and employment change for the corresponding county and industry.

3.3.2. Feature engineering with LLMs

To evaluate whether LLM-generated perceptual features can enhance online job fraud detection, two experimental groups were designed using ChatGPT (ChatGPT — Release Notes, 2026). A temperature setting of 0.1 was used for all API calls to ensure reproducibility across both experimental conditions.

In the first experimental group, a single-pass API call approach was used to obtain ChatGPT-4o's perceived fraud risk for each job posting in the EMSCAD. As illustrated in Fig. 3, the prompt instructs ChatGPT to act as a job fraud analyst and return a single integer score ranging from 0 to 10, where 0 indicates the lowest perceived fraud risk and 10 indicates the highest. Each prompt was processed independently, with no memory retention, tool use, or multi-step reasoning involved. The prompt language can be found in Appendix A.

In the second experimental group, we implemented a multi-agent framework through task decomposition and specialization (He et al., 2025; Sapkota et al., 2026). The Multi-Agent Framework provides greater tractability and explainability for fraud risk perception, increasing the opportunity for future refinement (Trad & Chehab, 2026). This design is informed by hierarchical multi-agent frameworks in which an orchestrator coordinates multiple specialized agents (Hosseini & Seilani, 2025; Zhao et al., 2022).

As shown in Fig. 4, the system consists of one Orchestrator Agent, one Final Judger Agent, and three specialized Sub-Agents: the Consistency Agent, the Legitimacy Agent, and the Context Agent. Upon receiving a job posting, the Orchestrator Agent invokes each Sub-Agent and aggregates their outputs to invoke the Final Judger Agent. The Final Judger Agent produce a final fraud risk score from 0 to 10 based solely on the output of the three Sub-Agent, without accessing the original job posting data. This ensures that the multi-agent framework produces risk scores by following a domain knowledge informed workflow. In addition to the risk score, the system generates a concise summary of detected red flags associated with online employment fraud. The flag definitions and detection guidelines can be found in Appendix B. The flag interpretation guidelines for Final Judger Agent can be found in Appendix C.

Each Sub-Agent independently perform a specialized task. The Consistency Agent examines internal coherence across job attributes such as geographic location, qualifications, job requirements, and compensation. Inconsistencies such as mismatched workplace locations and company addresses may signal fraud, because deceptive postings often prioritize persuasion over informational accuracy (Gong et al., 2025; Hou et al., 2019; Mbaziira et al., 2022; Nanath & Olney, 2023).

The Legitimacy Agent identifies four common red es: missing information, unrealistic compensation, sensitive information requests, and pressure tactics. Missing information includes absent or incomplete fields such as company names. Unrealistic compensation is characterized by unusually high wages paired with minimal qualifications or effort (Hou et al., 2019). Sensitive data requests and pressure tactics often co-occur, with offenders creating urgency or fear to induce compliance (Lazarus et al., 2023).

The Context Agent evaluates external consistency by collecting and analyzing data related to the job's stated location and industry. Specifically, it compares advertised salaries with county-level industry averages, assesses local industry size and employment trends, and verify if employer name is clearly stated. External data sources include the 2013 QCEW dataset (see Sections 3.2 and 3.3). Job postings aligned with local labor market conditions and explicitly stated the employer name are assumed to present lower fraud risk (Armstrong & Forde, 2003).

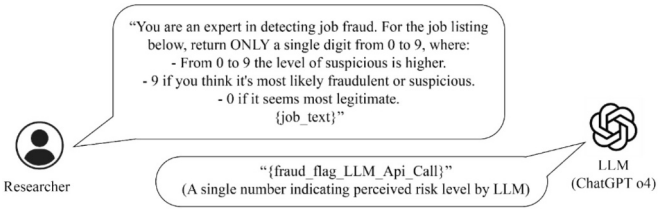


Fig. 3. Typical query prompt and responses of single-pass API call.

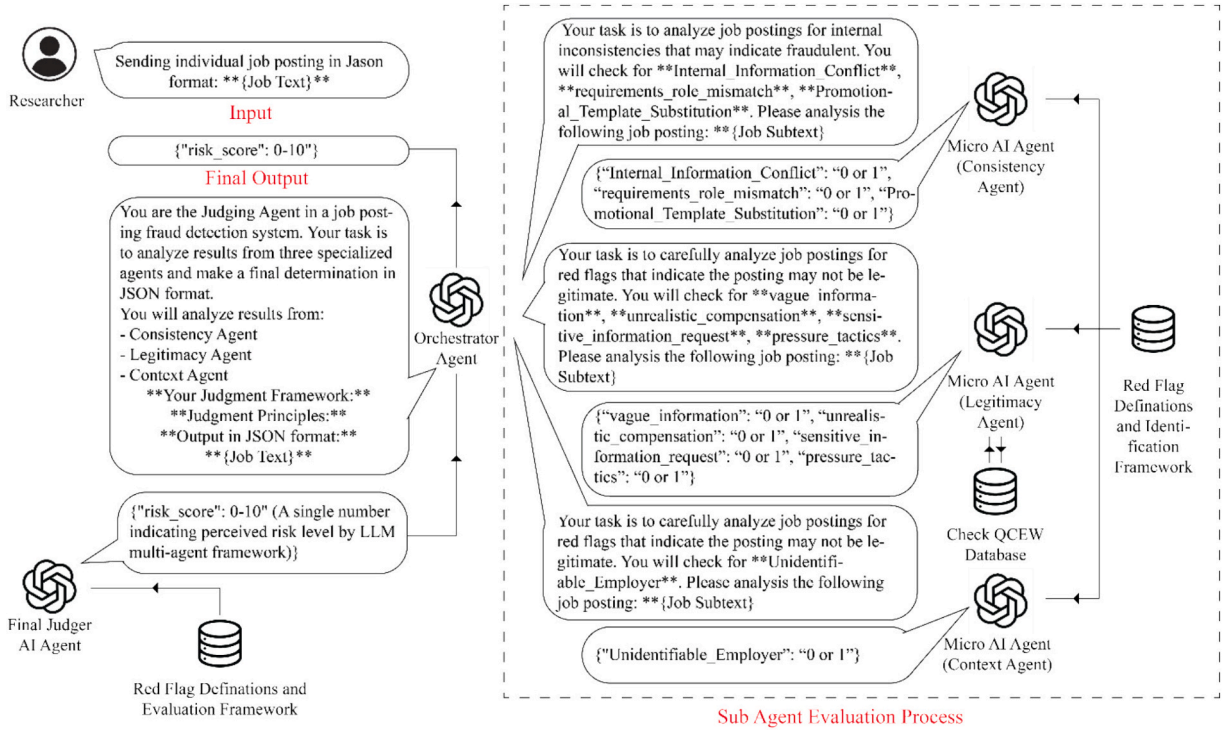


Fig. 4. Structure of multi-agent framework.

3.4. Machine learning models

3.4.1. Model selection and data preparation

Risk scores generated by both the single-pass API call approach and the multi-agent framework (Section 3.3) were incorporated as additional features in ML models. Model performance with and without these risk scores was compared to assessing whether LLMs capture fraud-related signals beyond those detected by traditional methods. Performance gains attributable to LLM-generated features indicate the presence of informational cues overlooked by conventional models.

This study assessed four ML models, which are LR, RF, XGBoost (XGB), and Multilayer Perceptron Neural Network (MLP). These models have been adopted by numerous previous fraud detection studies for their strong performances and ease to use (Gong et al., 2025; Hosseini & Seilani, 2025; Röhner et al., 2022). While LR provides a simple, interpretable baseline model, RF and XGB are nonlinear ensemble model that works well on binary classification tasks (Calanna et al., 2020; Röhner et al., 2022). MLP and other Neural networks have been increasingly adopted in recent years due to their superior accuracy on prediction tasks involving imbalanced datasets (Karnavou et al., 2025). Table 1 compares the characteristics of these four models.

Table 1
Machine learning models and data processing processes summary.

Model	Linearity	Model Family	Text Data Processing	Numeric Data Processing
Logistic Regression (LR)	Linear	Regression based	Dense TF-IDF	Standardized
Random Forest (RF)	Nonlinear	Tree based	Dense TF-IDF	Raw
XGBoost (XGB)	Nonlinear	Tree based	Sparse TF-IDF	Raw
BERT-Multilayer Perceptron Neural Network (MLP)	Nonlinear	Neural Network	BERT CLS Embedding	Standardized

This study uses Frequency–Inverse Document Frequency (TF–IDF) and Bidirectional Encoder Representations from Transformers (BERT) for text feature extractions. TF-IDF converts textual content into numerical representations based on word importance within the EMSCAD corpus. The TF-IDF vectorizer computes numerical values for each word based on its frequency within the EMSCAD dataset, returning a sparse matrix where each vector representing a word with the values indicating word importance (Bozkurt et al., 2024; Usman et al., 2024). The dimensionality of the TF–IDF matrix was controlled through the word limit parameter, which is set to 5000 for the Sparse TF-IDF and 300 for the Dense TF-IDF (Potshangbam & Singh, 2026; Shuang et al., 2018; Yousefi-Azar & Hamey, 2017; Zhang et al., 2023). Sparse TF–IDF representations were applied to XGBoost (XGB), which is optimized for sparse inputs, while dense TF–IDF was used for logistic regression (LR) and random forest (RF).

A pretrained BERT Base Uncased model is used to establish a MLP baseline. BERT is a transformer-based language model developed by Google and designed for deep bidirectional contextual understanding of natural language (Devlin et al., 2019). Following the conventional approaches developed in previous studies, we leveraged the special classification token (CLS) in BERT to absorb contextual meanings of text inputs features, including Description, Requirements, Benefits, and Company Profile. The final activation vectors of CLS were combined with numeric, Boolean, and categorical features and passed to a MLP with two hidden layers. Similar to other machine learning models, we compared BERT MLP baseline model with single-pass LLM enhanced model and multi-agent enhanced model using Monte-Carlo Cross Validation with random seed (from 1 to 50 in each iteration).

Categorical variables were encoded using one-hot encoding for the 20 most frequent categories, with less frequent and missing values grouped into an "other" category, a strategy commonly adopted to balance representativeness and generalizability (Duque et al., 2025; Jia et al., 2025; Shehab et al., 2025). Numeric variables were standardized for LR and MLP, while RF and XGB used unscaled values due to scale invariance (Baral & Haq, 2020). Missing numeric values were imputed using mean substitution (see Table 2).

Table 2
Summary of variables used in machine learning model training and evaluation.

Column	Description	Variable Type	Source	Number of Missing Values
Fraudulent (Target Variable)	Target classification attribute	Boolean	EMSCAD	0
Company Profile	A brief company description	Text	EMSCAD	1517
Requirements	Enlisted requirements for the job opening	Text	EMSCAD	1389
Benefits	Enlisted offered benefits by the employer	Text	EMSCAD	3823
Description	Detailed description of the job ad	Text	EMSCAD	0
Department	Corporate department (e.g., sales)	Text	EMSCAD	6279
Location	Geographical location of the job ad	Text	EMSCAD	0
Title	The title of job ad entry	Text	EMSCAD	0
Required Experience	Executive, Entry level, Intern, etc.	Categorical	EMSCAD	3455
Required Education	.Bachelor, master's degree, doctorate	Categorical	EMSCAD	3544
Function	Consulting, Engineering, Research, Sales, etc.	Categorical	EMSCAD	3114
Industry	Automotive, IT, Health care, Real estate, etc.	Categorical	EMSCAD	2088
Employment Type	Full-time, Part-time, Contract, etc.	Categorical	EMSCAD	1382
Telecommuting	True for telecommuting positions	Boolean	EMSCAD	0
Has Company Logo	True if company logo is present	Boolean	EMSCAD	0
Has Questions	True if screening questions are present	Boolean	EMSCAD	0
Salary Range Lower	The lower range of the indicative salary range (e.g., 50,000 from $50,000–$60,000)	Integer	EMSCAD	7523
Salary Range Upper	The upper range of the indicative salary range (e.g., 60,000 from $50,000–$60,000)	Integer	EMSCAD	7523
Single-Pass API Call Risk Score	The perceived risk score generated by single-pass LLM API call (range 0–10)	Integer	ChatGPT-4oAPI Call	0
Multi-Agent Framework Risk Score	The perceived risk score generated by LLM multi-agent framework (range 0–10)	Integer	ChatGPT-4oAI-Agent System	0

Note: Length of EMSCAD subset used for this study: 8781 (9099 unsurpassable rows removed, see Section 3.3.1). Length of original EMSCAD dataset: 17,880.

The analysis used an imbalanced subset of the EMSCAD dataset consisting of 8781 entries, including 556 fraudulent cases. To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data only (Chawla et al., 2002), while test sets retained the original class distribution to prevent data leakage. SMOTE is widely adopted for binary classification tasks with imbalanced data (Roy et al., 2025; Yeşilkanat & Akkoyun, 2025).

3.4.2. Model performance evaluation

We adopted stratified Monte Carlo cross-validation, which is also known as repeated holdout, for all model evaluations and comparisons (Esengönül et al., 2025). This method is particularly suitable for datasets with class imbalances (Xu & Liang, 2001; Zeng et al., 2024). For each model, we performed 50 iterations of stratified 80/20 random splits. At each iteration, three feature groups were used as input to train the ML model: (1) EMSCAD data only (baseline), (2) EMSCAD data plus single-pass API call risk scores, and (3) EMSCAD data plus multi-agent framework risk scores.

We selected accuracy, recall, F1 score, and Precision-Recall Area Under the Curve (PR-AUC) as the model performance metrics. They were calculated at each iteration to enable paired *t*-tests. Accuracy measures overall correctness, recall captures sensitivity to fraudulent cases, F1 and PR-AUC summarize overall performance balancing accuracy and recall.

The code and data of this study is uploaded to a GitHub Repo, which can be identified in **Code Availability Statement** section.

4. Results

4.1. LLMs detection of fraudulent job postings

The single-pass API call and the multi-agent framework were evaluated for job posting fraud detection. Single-pass API call required an average of 5–7 s per posting, while the unoptimized multi-agent framework required 30–40 s; parallel processing using multithreading reduced this to 15–20 s. The token consumption speed for single-pass API call is approximately one-third of multi-agent framework. Risk scores generated by direct single-pass API call exhibited a trimodal distribution, clustering at low (0–2) and medium (3–4) to high (5–9) scores (Fig. 5A). Fraud prevalence increased with risk scores (Fig. 5B), rising from 1.78% at score 0 to 77.05% at score 9.

4.2. Feature engineering performance enhancement

Feature engineering is central to ML model training, as the choice and quality of features can substantially affect model performance (Chia et al., 2021; Reid Turner et al., 1999; Wang et al., 2021). High quality engineered features can significantly improve the model performances (Richardson et al., 2024), whereas redundant and irrelevant information may introduce noise and increase overfitting (Lei et al., 2025).

As shown in Table 3, all four models (LR, RF, XGB, and MLP) improved on at least one performance metric with LLM-generated features, with no significant performance degradation observed. The multi-agent approach significantly improved two top-performing metrics (Cohen's d between 0.364 and 0.452) while the single-pass LLM API approach improved three (Cohen's d between 0.397 and 0.674). False negative cases (measured by recall), resulting in undiscovered fraudulent cases that impose monetary and mental costs to victims, were significantly reduced by the single-pass API call approach for LR and RF (Cohen's d 0.553 and 0.674).

Accuracy gains were significant for LR and RF despite the imbalanced nature of the dataset. The F1 scores are improved by single-pass API call in RF and by multi-agent framework in LR and XGB. PR-AUC saw improvements in RF, XGB and MLP. Overall, single-pass approach and multi-agent framework demonstrated similar performance, achieving significant improvements in 7 of 16 and 5 of 16 performance metrics respectively.

Difference values calculated as first group minus second group (e.g., LLM API vs. Baseline = LLM API - Baseline). Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

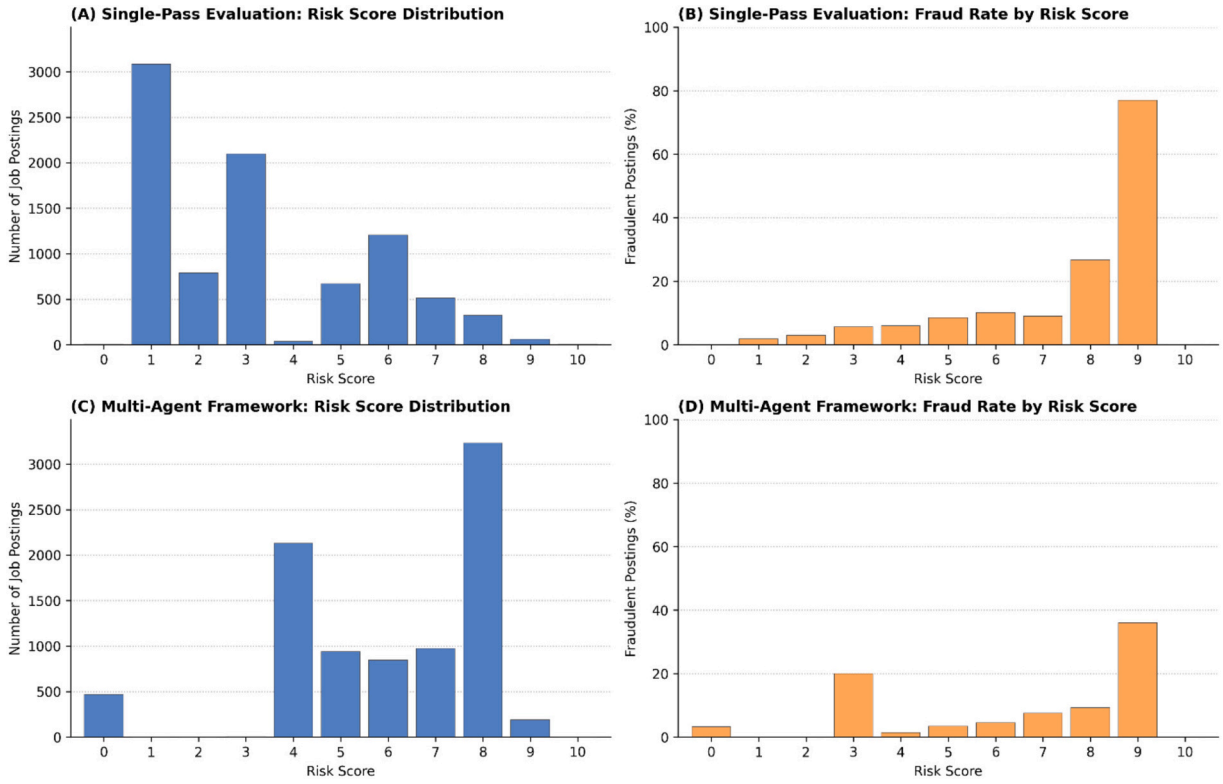


Fig. 5. Fraud rate and risk score comparison: single-pass API call vs. multi-agent framework. Panels A and C show the distribution of risk scores (0−10) for single-pass API call and multi-agent framework. Panels B and D show the percentage of postings that are fraudulent at each score. Note: For the single-pass API call, high-risk advertisements (scores 5–9) had, on average, a fraud rate of 12.91% compared to 2.01% fraud rate in low-risk advertisements (scores 0–2). On the other hand, the result obtained using the multi-agent framework shows a trimodal distribution thats shifted toward the high end of the scale, with separation between low (0–2), mid (3–6) and high (7–9) risk scores (Fig. 5C). The high fraud rate for risk score 3 appears to be an ignorable outlier with only 5 postings and 1 fraudulent case (Fig. 5D). Both single-pass API call and multi-agent framework show monotonic correlation between risk scores and fraud rate, with single-pass API call slightly outperform (Spearmans $\rho = 0.207$, $p < 0.001$) *multi-agent framework (Spearmans* $\rho = 0.170$, $p < 0.001$*) in risk score/fraud rate alignment.*

Table 3
Machine learning model performance comparison between Baseline, LLM API call enhanced, and multi-agent system enhanced approaches.

Model	Metric	Baseline	LLM API vs Baseline				Multi-Agent vs Baseline			
				95% CI				95% CI		
			Difference	Lower	Upper	Cohen's d	Difference	Lower	Upper	Cohen's d
LR	Accuracy	0.969	0.000	−0.001	0.001	−0.075	0.001	0.000	0.001	0.281
	Recall	0.865	0.010***	0.006	0.015	0.674	0.003	0.000	0.007	0.250
	F1	0.777	0.001	−0.003	0.005	0.080	0.004*	0.000	0.008	0.300
	PR-AUC	0.899	0.001	0.000	0.003	0.220	0.001	0.000	0.003	0.252
RF	Accuracy	0.980	0.001**	0.000	0.001	0.397	0.000	0.000	0.001	0.234
	Recall	0.709	0.012***	0.006	0.019	0.554	0.005	−0.001	0.012	0.233
	F1	0.819	0.007**	0.003	0.011	0.454	0.004	−0.001	0.009	0.234
	PR-AUC	0.918	0.003**	0.001	0.005	0.406	0.000	−0.002	0.002	0.014
XGB	Accuracy	0.980	0.001	0.000	0.001	0.256	0.001**	0.000	0.002	0.452
	Recall	0.771	−0.002	−0.008	0.005	−0.072	0.000	−0.006	0.007	0.016
	F1	0.831	0.003	−0.002	0.008	0.191	0.007*	0.001	0.012	0.364
	PR-AUC	0.906	0.004*	0.001	0.007	0.370	0.004*	0.001	0.006	0.361
MLP (BERT)	Accuracy	0.974	0.001	0.000	0.001	0.190	0.000	0.000	0.001	0.151
	Recall	0.754	0.006	−0.003	0.016	0.204	0.008	−0.001	0.017	0.260
	F1	0.783	0.005	−0.001	0.011	0.233	0.005	−0.002	0.011	0.208
	PR-AUC	0.842	0.008**	0.003	0.013	0.418	0.006*	0.001	0.011	0.353

Note: 95% CI stands for 95% confidence interval of the difference.

4.3. Multi-agent reasoning assessments

The design principle of our multi-agent framework is centered around the assimilation of fraud detection domain knowledge, into specialized sub-agents. Each specialized agent was designed to detect given red flags identified through a literature review of criminology studies. Specifically, the legitimacy AI agent targets four red flags, which are sensitive information requests (Lazarus et al., 2023), unrealistic compensation (Hou et al., 2019), pressure tactics (Quah & Sriganesh, 2008; Shang et al., 2022), and vague information (Costello & Lee, 2022). The context AI agent evaluates unidentifiable employer flag (Armstrong & Forde, 2003). The consistency agent identifies requirement's role mismatches (Hou et al., 2019), location conflicts (Gong et al., 2025), and content overlap (Mbaziira et al., 2022; Nanath &

Olney, 2023).

Table 4 summarizes the fraud detection performance of individual red flags. To validate the reliability of AI-generated flags, we conducted a manual validation experiment with a subsample of 50 job postings, which was generated using stratified random sampling strategy to ensure balanced representation of flags. A human coder reviewed and coded the flags for the subsample without access to the AI sub-agents results. By comparing the results of human coder and AI, we found fair to near perfect inter-rater agreement with Cohen's k ranging from 0.308 to 0.922. Full results of the validation experiment are provided in Appendix D. These findings indicate that the flags generated by the multi-agent system are mostly reliable.

Relative risk indicating the multiplicative increase in fraud probability when a given flag is present. Sensitive Information Request was the strongest individual predictor (relative risk = 3.69) but had limited impact due to low prevalence. Unidentifiable Employer combined high predictive strength (relative risk = 3.67) with high prevalence (48.17%), making it a particularly effective indicator.

From the faking theory perspective, Consistency Agent flags, such as Location Conflict and Requirements Role Mismatch, reflect comprehension stage faking behavior in which fraudsters manipulate the information as they enter required information following the templates provided by online employments platforms. Context Agent flags reflect retrieval stage faking where fraudsters retrieve and distort memories to deliberately withhold verifiable details. Legitimacy Agent flags, including unrealistic compensation and sensitive information request, match the judgment and response stage faking behaviors where fraudsters deploy calculated signals designed to maximize victim compliance and trust. In earlier cognitive psychological models, faking behaviors were identified to only take place at the response stage (Tourangeau & Rasinski, 1988). In our study, the strong predictive power of Legitimacy Agent and Context Agent flags suggests that faking behaviors also occurred at comprehension and retrieval stages, which align with findings of more recent studies (Röhner et al., 2025).

Table 5 illustrates a very weak but statistically significant monotonic relationship between the number of concurrent red flags and fraud rate (Spearman's ρ: 0.152, $p < 0.001$). The fraud rate increased 31.3-fold from 3.2% for postings with no flags to 100% for postings with seven concurrent flags. The increment in fraudulent prevalence rate was gradual with fewer than 4 concurrent flags. However, the fifth to seventh red flag substantially increased the fraudulent prevalence rate from 8.52% to 32.26%, 75.56%, and 100%, respectively. This trend aligns with findings from prior research that fraudulent job postings are commonly associated with multiple concurrent red flags (Hou et al., 2019; Nanath & Olney, 2023b).

However, two distinct challenges are observed if a threshold-based detection system were implemented based on number of concurrent red flags. First, postings with 0 to 1 concurrent red flag account for approximately 32.0% of all fraudulent cases in the dataset, yet their fraudulent prevalence rates remain between 2.05% and 3.2%. Second, at the 2 to 4 concurrent flags group, fraudulent prevalence remains 5.95% to 8.52%, indicating that the majority of postings at this level are legitimate and generate substantial false positive predictions in a threshold-based detection system. These observations suggest that the number of concurrent red flags alone is an insufficient basis for risk classification. Instead, number of concurrent flags should be carefully calibrated in practice and differences in predictive power of individual flag should also be considered.

Table 5
Fraud detection rate by number of concurrent red flags.

Number of Concurrent Red Flags	Total Postings	Fraudulent Postings	Fraudulent Prevalence Rate
0	469	15	3.2%
1	2337	48	2.05%
2	3095	184	5.95%
3	1990	169	8.49%
4	716	61	8.52%
5	124	40	32.26%
6	45	34	75.56%
7	5	5	100%

Note: The Spearman correlation test shows a very weak ($\rho = 0.152$) but statistically significant ($p < 0.001$) monotonic relationship between number of concurrent red flags and fraudulent prevalence rate.

5. Discussion

This exploratory study investigates the application of LLMs and AI Agents in cyber employment fraud detection and prevention. Three major findings emerge. First, the risk scores engineered by both single-pass approach and multi-agent framework significantly improved four traditional ML algorithms' performances (RQ1). The multi-agent system achieved statistically significant improvements in 5 of 16 key performance metrics while single-pass LLM API calls improved 7 metrics. Notably, no performance metrics statistically decreased Second, the single-pass API call was more cost effective in time and token consumptions while showing a comparable performance to multi-agent framework. The multi-agent framework generates risk scores by strictly following a domain-knowledge-informed workflow, offering greater tractability, explainability, and human control. The design of Multi-Agent Framework ensures compliance with certain regulations, including EU Artificial Intelligence Act (AI Act) which set explainability as a legal requirement for high-risk AI (van Beem, 2026) (RQ2). Third, substantial variation in red flags' effectiveness, where relative risks ranged from 0.86 to 3.69 by flag, was observed (RQ3). From a criminological perspective, this pattern is consistent with cumulative risk frameworks, and the convergence of multiple situational warning signs substantially elevates the likelihood of victimization. Taken together, these findings demonstrate that both single-pass approach and criminology-informed multi-agent framework system can jointly strengthen cyber employment fraud detection in platform-based environments.

LLMs can theoretically significantly advance deception detection through their sophisticated natural language processing capabilities. However, the previous applications of LLMs in fraud detection mainly

Table 4
Fraud detection performance by individual red flag type.

Flag Type	Detected By	Count	Flag Prevalence Rate	Fraudulent Rate with Flag	Fraudulent Rate without Flag	Relative Risk
Sensitive Information Request	Legitimacy Agent	39	0.44%	23.08%	6.26%	3.69***
Unidentifiable Employer	Context Agent	4230	48.17%	10.17%	2.77%	3.67***
Requirements Role Mismatch	Consistency Agent	1935	22.04%	12.66%	4.54%	2.79***
Unrealistic Compensation	Legitimacy Agent	806	9.18%	14.27%	5.53%	2.58***
Pressure Tactics	Legitimacy Agent	520	5.92%	12.31%	5.96%	2.07***
Contents Overlap	Consistency Agent	2396	27.29%	8.64%	5.47%	1.58***
Vague Information	Legitimacy Agent	7455	84.9%	6.52%	5.28%	1.23
Location Conflict	Consistency Agent	905	10.31%	5.52%	6.42%	0.86

Note: The differences of fraudulent rates in groups with and without flags are tested with Fisher's exact test (Sensitive Information Request) and Chi Square test (all other flags). Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

focused on synthetic data generation and Domain Generation Algorithms (DGAs) detection (Goyal & Mahmoud, 2025; Ouesleti et al., 2024; Pelayo-Benedet et al., 2025; Tubishat et al., 2025). A systematic literature review revealed limited usage of LLM and text mining in fraud detection and risk management domain (Kumar et al., 2021). This study experimented with usefulness of LLMs in fraud detection and risk management by collecting the risk perception of LLMs on EMSCAD datasets via both a direct single-pass API call and a more complicated multi-agent system framework (see Fig. 5).

We observed a very weak to weak monotonic but statistically significant relationship between the fraud rate and risk scores generated by single-pass LLMs API call and multi-agent system (Spearman's correlation ρ of 0.207 and 0.170 respectively). The fraudulent prevalence rate in low risk groups ranged from 0% to 2.92% for single-pass API call and 3.20% for the multi-agent framework, dropping from 6.3% average fraudulent prevalence rate for the entire dataset. However, the risk group represents a small fraction of the entire dataset, especially for single-pass API call (3876 postings) and the multi-agent framework (469 postings). Relying solely on LLMs or other complex architectures may not be sufficient to address the challenges posed by class imbalance and data irregularities in fraud detection and similar applications (de Carvalho et al., 2025). From a criminological standpoint, this limitation underscores the importance of embedding domain knowledge into automated systems, as statistical pattern recognition alone cannot fully handle the complex interactions between human, social, and technical factors.

While the single-pass API call is cost-effective in risk perception, it may lack the explainability and tractability required in high-stake applications such as cyber employment fraud detection (Sreedharan et al., 2022; Zhang et al., 2022). Instead of solely relying on LLMs to make final judgment, this study proposes the integration of LLMs as feature engineering tools to enhance the performance of traditional fraud detection models. Our approach is similar to previous studies which employed social sensing and crowdsourcing techniques by collecting risk perceptions from human evaluators and use them as features to train ML models (Nanath & Olney, 2023; Nwaiwu et al., 2025). Our study extends beyond the findings of these previous studies by demonstrating that the features engineered by LLMs can achieve similar effects in ML model enhancements. With multi-agent framework and single-pass API call achieving significant improvements on 5 of 16 and 7 of 16 critical performance metrics respectively, our study reaffirms that LLMs can engineer useful features by implicitly learning signals and patterns (Nwaiwu et al., 2025). This finding reduces the intensive manual labor that is previously required by social sensing and crowdsourcing approaches in fraud detection and risk management domain.

Previous studies have explored the multi-agent system application in biomedical applications (Yang et al., 2025), manufacturing systems (Lee et al., 2025), and smart energy facility inspections (Cui et al., 2025). In these cases, the multi-agent systems often achieved statistically significant improvements in accuracy, interpretability and robustness to single LLM baseline. However, few previous studies investigated the adoption of multi-agent systems in online employment fraud detection, or more broadly, cybercrime prevention.

In this study, we propose a structured multi-agent system to decompose fraud detection into specialized, manageable subtasks handled by three separate subagents that specialize in information consistency, contextual alignment, and contents legitimacy. Our design achieved significantly better performances on fraud detection and feature engineering for traditional ML model enhancements. This finding demonstrates multi-agent system's advantage in discovering the hidden patterns in online job posting that traditional ML models often overlooked. This finding is consistent with a previous fraud detection experiment, where researchers found that auditors correctly identify fraud only when assessing multiple fraud cues together (Norman et al., 2010), which resembles the architecture of our multi-agent system.

It is noted that since our design has flexible assignments of subagents, future work could explore new subagents design by adjusting evaluation strategies.

6. Conclusion and implications

6.1. Theoretical implications

The findings of this study carry theoretical implications on both agentic AI and criminology. From an agentic AI perspective, the multi-agent framework operationalizes the virtual guardianship concept with three specialized sub-agents that independently evaluate eight fraud-related red flags of a job posting. Unlike ensemble methods that combine multiple models performing the same task (Goh et al., 2024), this concurrent validation design embeds criminology informed domain knowledge and enables a systematic quantitative assessment of the relative importance of individual red flags within an AI-based detection framework.

The results reveal substantial variation in the predictive utility of different red flags. Requests for sensitive information demonstrate the strongest predictive power, with a relative risk of 3.69, but contribute less overall due to their low prevalence of 0.44%. In contrast, contents overlap exhibits weak predictive strength, with a relative risk of 1.58, yet occurs frequently, appearing in 27.29% of all postings. This pattern is consistent with principles from information theory, which suggest that rarer indicators tend to carry greater informational value than common but less distinctive signals (Kim et al., 2025; Tiwary et al., 2024). From a criminology perspective, this predictive power of red flags in cyberspace mirrors the concepts of signal crime, where the occurrence of certain incidents is viewed as warning signals about the levels of risk exposure (Innes, 2004). Therefore, targeted monitoring of certain red flags is recommended. For example, unrealistic compensation emerges as an optimal signal because it combines relatively strong predictive power with moderate prevalence.

The multi-agent framework bears a meaningful alignment with the deception and faking behavior literature. The decomposition of fraud detection into consistency, context, and legitimacy sub-agents mirrors the heterogeneous nature of faking strategies described by Röhner et al. (2025), in which deceptive behavior operates in comprehension, retrieval, judgment, and response stages with various combinations rather than a single homogeneous act. Fraudulent job postings can be understood as a form of motivated information distortion (Bensch et al., 2019), in which actors strategically manipulate signals of organizational identity, compensation, and urgency to lure victims.

The findings further underscore the importance of evidence triangulation in cyber fraud detection. The co-occurrence of multiple red flags is associated with a 31.3-fold increase in the probability of fraud, indicating that different indicators capture distinct dimensions of fraudulent behavior. From a criminal justice perspective, this supports layered and multi-source detection strategies that reduce information asymmetry and strengthen guardianship in digital environments. Overall, the results suggest that the relationship between Agentic AI and criminology theory is integrative rather than merely additive. Embedding domain knowledge into the architecture of multi-agent framework can produce detection gains that traditional machine learning models cannot achieve, offering opportunities to further improve detection accuracy and support proactive cybercrime prevention.

6.2. Practical implications

Recent studies have highlighted the emergence of “dark LLMs,” referring to language models that are exploited by cybercriminals as tools for deception and fraud (Marulli et al., 2024; Shaikh et al., 2025). In contrast, little attention has been given to how LLMs can be deployed by criminal justice agencies and platform operators for prevention and detection (Kumar et al., 2021). This study addresses that gap by demonstrating that LLMs generated fraud risk scores can enhance the

performance of traditional predictive models, including logistic regression, random forest, XGBoost, and multilayer perceptron models. These results indicate that LLM based systems can serve as operational decision support tools for law enforcement, regulators, and private sector platform operators engaged in fraud detection.

For online employment platforms such as LinkedIn and Indeed, our work proves the feasibility of integrating LLMs for proactive guardianship. Modern LLMs and criminology informed multi-agent frameworks have the potential to serve as a perceptual layer that augments existing detection pipelines or deployed as an independent risk screening mechanism to flag suspicious job postings before victimization occurs. In this study, postings with five or more concurrent red flags, where fraudulent prevalence exceeds 30%, represent high predictions that account for approximately 14.2% of all fraudulent cases in the dataset and are strong candidates for automated fraudulent screening. For cases with fewer concurrent red flags, human-in-the-loop review is warranted to manage the precision-recall tradeoffs inherent to the difference in predictive power of various red flags. Note that EMSCAD is a relatively old dataset. The contemporary deployment of LLMs integrated system would require calibration using more recent fraudulent cases.

From a criminal justice perspective, this study demonstrates LLMs' potential in enabling earlier intervention, reducing investigative burden, and supporting coordinated action between platform providers and enforcement agencies. The design principles of specialized sub agents, domain informed indicators, and concurrent validation are transferable to other forms of cyber enabled fraud, including financial scams, identity theft, and marketplace deception, with appropriate domain specific adaptation.

The study also advances explainable and accountable AI for criminal justice practice. Each red flag is grounded in criminological theory, allowing risk assessments to be transparent, interpretable, and auditable by human analysts. This supports evidentiary review, enhances procedural fairness, and enables analysts to identify emerging fraud patterns across cases and jurisdictions. By embedding interpretability into system design, the framework facilitates trust, accountability, and policy oversight in the deployment of AI enabled cybercrime prevention tools.

6.3. Policy implications

This study highlights several policy implications for criminal justice agencies and regulators concerning the use of LLMs to detect and prevent cyber employment fraud. By using EMSCAD dataset, we found that ChatGPT-4o is capable of automatically generating useful features for risk sensing and fraud detection. Such capabilities could potentially address a persistent policy challenge, namely that traditional reactive enforcement cannot keep pace with rapidly evolving online fraud schemes.

From a resource allocation perspective, the results also indicate important tradeoffs between detection accuracy and computational cost. Multi-agent framework provides stronger tractability but require longer processing times than single-pass API call approaches. Policymakers and agency leaders should therefore consider tiered deployment strategies, in which resource intensive multi-agent analysis is reserved for high-risk cases, while faster screening tools are used for large scale initial monitoring. Such an approach allows agencies with limited computational infrastructure to adopt LLMs assisted detection without overextending operational capacity.

More broadly, the findings underscore that cyber employment fraud is not solely a technical problem but a governance and regulatory challenge that requires interdisciplinary coordination. LLMs assisted prioritization can support fair and efficient allocation of investigative resources by directing attention to high harm and high-risk cases, thereby improving the effectiveness of limited enforcement capacity. At the same time, the deployment of AI in criminal justice contexts raises critical issues related to transparency, accountability, bias, and due process. The domain informed design of the multi-agent framework, in which each red flag is grounded in criminological theory, supports interpretable and auditable decision making that aligns with procedural justice requirements.

Accordingly, policymakers should consider establishing regulatory standards for AI-assisted fraud detection that require explainability, human oversight, and auditability of automated risk assessments. Such standards would enhance accountability and facilitate judicial review of platform level enforcement actions. In addition, policy frameworks should encourage structured data sharing and collaboration among law enforcement agencies, regulatory bodies, platform providers, and academic researchers to validate, monitor, and continuously improve LLMs based fraud detection systems. Collectively, these policy measures can strengthen institutional capacity to prevent cyber employment fraud, enhance platform accountability, and support coordinated cross jurisdictional responses to digitally mediated crime.

6.4. Limitations and future directions

This study has several limitations that point to important directions for future research. First, the EMSCAD dataset was collected between 2012 and 2014 and may not fully capture the most recent tactics used in cyber employment fraud (Vidros et al., 2017). Although core deception mechanisms likely persist, recent evidence indicates that offenders continue to evolve their strategies by leveraging advanced technologies such as blockchain systems, large scale data analytics, and autonomous tools (Chiang et al., 2025; Richet, 2022; Uddin, 2025). Future research should validate the proposed multi-agent framework using more recent and operational datasets to assess its robustness, temporal stability, and generalizability to contemporary fraud patterns. As online employment fraud patterns continue to evolve, it is strongly recommended that new red flags and red flag combinations be tested under the multi-agent framework.

Second, the design of the multi-agent architecture was theory driven and exploratory rather than empirically optimized. The number and configuration of red flags were not systematically varied, and the three agent structures consisting of consistency, legitimacy, and context assessment was specified a priori. As an initial probe of this design, we conducted a leave-one-out ablation in which each sub-agent was removed in turn and the Final Judging Agent re-computed risk scores from the remaining two agents' flags, following the same workflow as the full system. The result of this ablation analysis can be found in Appendix E. Every two-agent configuration retained a statistically significant monotonic relationship with the fraud rate, though weaker than that of the full three-agent system. In the Monte Carlo cross-validation, removing a sub-agent reduced performance for most classifiers but slightly improved it for XGBoost, suggesting that three-agent framework may not be optimal for every classifier and that the optimal configuration may be model-specific. Future studies should conduct systematic experiments and hyperparameter tuning to evaluate alternative configurations, including different numbers of sub agents, alternative domain informed indicators, and task specific agent roles. Such analyses would clarify which components contribute most to detection accuracy and would support the development of more efficient and scalable system designs for operational use.

The three sub-agents in the current multi-agent framework were designed based on criminological domain knowledge and prior fraud detection literature. While the agent structure and red flag types retrospectively mirror the heterogeneous nature of faking strategies, our study did not incorporate individual level decision-making processes into the original research design. Future research should explore whether a multi-agent system explicitly designed around this faking taxonomy, in which each sub-agent is assigned to assess a specific stage of deceptive behavior, would achieve higher detection accuracy than the current design. Such research design may require a richer dataset capturing the behavioral process of posting construction, including platform interaction logs and revision histories, which are not available

in the EMSCAD dataset. Nevertheless, the integration of qualitative faking evaluation frameworks into agentic AI architectures represents a promising direction for advancing both the theoretical grounding and practical effectiveness of automated fraud detection systems.

Third, this study evaluated a single large language model through the ChatGPT-4o API. Although the model used represents a highly capable LLM, the findings may not generalize across other model architectures. Future work should compare performance across multiple proprietary and open source LLMs, including Claude, Gemini, LLaMA, and Mistral, to determine whether the observed gains reflect model specific characteristics or broader properties of LLM based feature engineering. Comparative studies should also examine tradeoffs between performance, computational cost, and deployment feasibility to inform resource allocation decisions for criminal justice agencies and platform operators.

Finally, future research should examine the real-world implementation of LLMs based fraud detection systems in collaboration with law enforcement agencies and online platforms. Field evaluations could assess how such systems affect investigative workflows, case prioritization, and victim protection outcomes, as well as potential risks related to bias, over flagging, and procedural fairness. Attention to governance, accountability, and human oversight will be essential to ensure that AI assisted detection systems support lawful, transparent, and equitable cybercrime prevention.

Despite these limitations, the findings provide evidence that multi-agent framework has strong potential for enhancing cyber employment fraud detection and represent a promising foundation for advancing AI enabled guardianship in the criminal justice system.

6.5. Conclusion

Addressing the challenges of unstructured text analysis and class imbalance in cyber fraud data, this study develops an explainable multiagent LLM framework that generates reliable fraud risk scores and identifies criminology informed red flags in job postings. To the best of our knowledge, this is the first LLM based feature engineering approach specifically designed to enhance the performance of traditional machine learning models in cyber employment fraud detection, offering an interpretable workflow that supports benchmarking, methodological development, and theory building within criminal justice research.

The proposed framework consists of three specialized sub agents that independently assess job posting consistency, legitimacy, and contextual signals using eight fraud indicators identified in prior research. Risk scores generated by the multi-agent framework were compared baseline models using Monte Carlo cross validation across four machine learning models including logistic regression, random forest, XGBoost, and multilayer perceptron networks.

The results show that LLM derived risk scores exhibit a significant monotonic relationship with fraud prevalence rate and improve model performance when incorporated as features. The multi-agent framework and singlepass API call approach showed similar performance in improving model metrics. The single-pass API call is more cost-effective while multi-agent framework provides greater tractability and human control. Analysis of red flags reveals substantial variation in predictive power, with relative risks ranging from 0.86 to 3.69, and demonstrates that the co-occurrence of multiple indicators increases the likelihood of fraud by a factor of 31.3. These findings confirm the value of domain informed, multi-source verification in detecting digitally mediated fraud.

This study makes three primary contributions. First, it demonstrates that LLMs can automate the extraction of fraud relevant features from unstructured text in ways that enhance traditional predictive models. Second, it shows that multi-agent framework and single-pass API call can jointly enhance the cyber employment fraud detection workflow. Third, it provides the first quantitative assessment of the relative effectiveness of fraud indicators within an AI driven detection framework.

From a criminal justice perspective, the findings highlight the growing importance of proactive, technology enabled guardianship in cyberspace. LLMs based systems can assist law enforcement agencies, regulators, and platform operators in identifying high risk content before victimization occurs, thereby shifting cybercrime control from reactive enforcement to preventive intervention. As cybercriminals increasingly leverage AI tools, future research should focus on optimizing multi-agent systems, evaluating their deployment in real world enforcement settings, and extending this approach to other forms of cyber enabled crime. Collectively, this work contributes to the development of scalable, transparent, and evidence-based strategies for cybercrime prevention in digitally mediated environments.

Code availability statement

The source code and replication scripts for this study are openly available in the folllowing GitHub repository: https://github.com/AnonymousAuthor333.

CRediT authorship contribution statement

Quan Sun: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ling Wu:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Xinyue Ye:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Claire Lee:** Writing – review & editing. **Wei Li:** Writing – review & editing. **Cuiling Liu:** Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors employed the OpenAI ChatGPT-5 model as a language-editing aide to enhance readability, grammar, and stylistic clarity. ChatGPT-5 was not used to create, expand, or alter the scientific ideas, data analyses, or conclusions presented herein, and no AI-generated text has been included verbatim in the manuscript. The authors used Claude for drafting and debugging the Python scripts used for data analysis. The generated code was reviewed, tested, and verified for accuracy by the authors, who take full responsibility for the integrity of the analysis.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcrimjus.2026.102692.

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